



Industrial PC

Debian 11 OS on RK3399

User Manual

For RK3399 Products

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Debian 11 OS

Debian 11 OS on RK3399 User Manual



This is the software manual for RK3399 Chipsee industrial PC. If you've never developed on this hardware with a Debian 11 OS, this manual can get you started quickly.

Backup Your OS Image For Bulk Installation

If you have finished developing your software, and plan to “copy” the whole system to many other Chipsee industrial PCs, you can backup the OS to an image file, just like the **.img** file you downloaded from Chipsee, or the OS we installed in the factory for you before shipping. And then you can flash it to many more devices.

Prepare for backup

We will use **SDDiskTool** to flash a bootable SD card, let your Chipsee PC boot from this SD card, then use this system to backup your OS image (the whole content on eMMC rootfs partition). You will need:

- **SDDiskTool** ([Click to download](#)).
- 16GB or larger micro SD card.
- SD card reader (to be used on your HOST PC).
- A Windows PC to run the SDDiskTool.
- A (X86 or X86_64) Linux HOST PC or virtual machine to make a new img file (make sure there is 25GB or more free space on the disk for the following process).
- Two **Chipsee prebuilt image**, one is the image that you are developing your software on, the other is a prebuilt-xxx-sd-xx.img, if you cannot find the prebuilt-xxx-sd-xx.img of your device, you can use just the prebuilt-xxx-emmc-xx.img temporarily, we will release the sd image later.

Note

More on the prebuilt image: the core idea of backup is to “swap” your data and the prebuilt data. So we will need to download a prebuilt image that you’re developing your software on (the OS image that you’re currently using on the Chipsee PC), unpack that image, swap the data, then repack the image.

Note

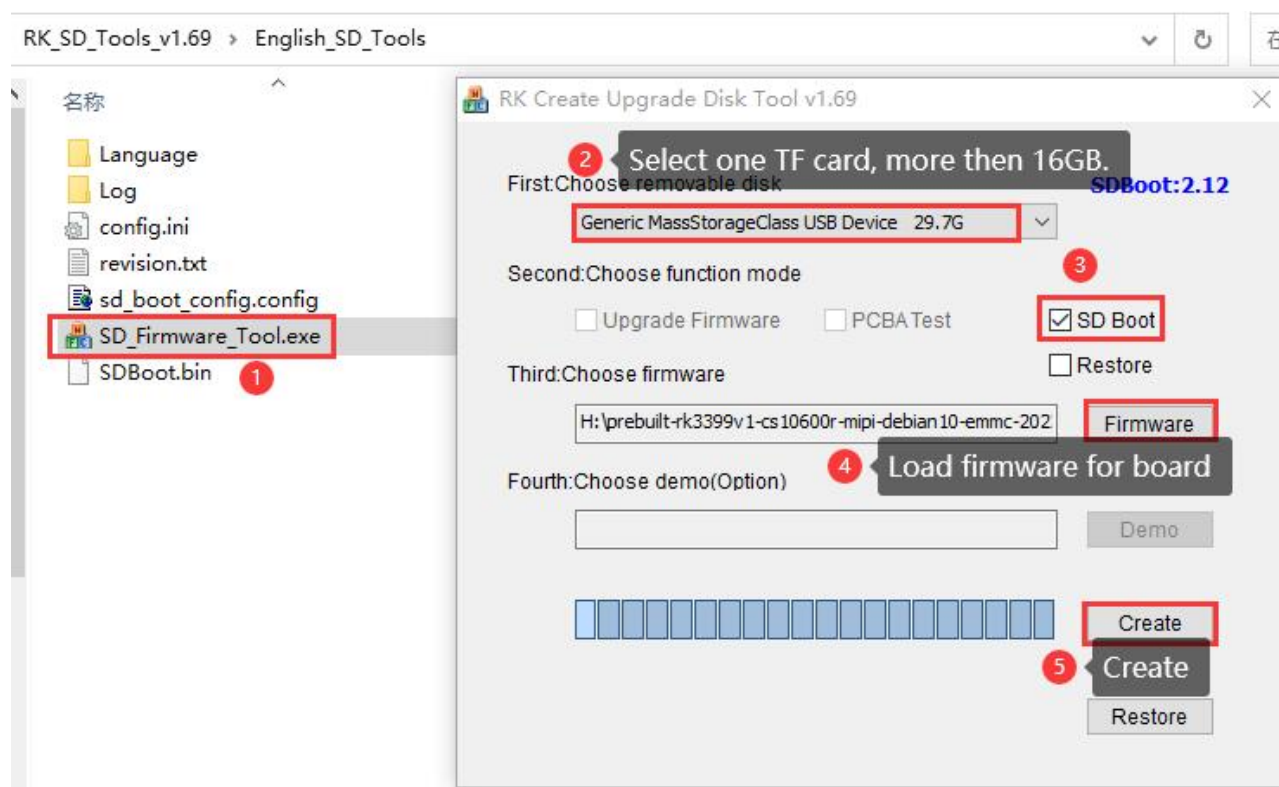
As for the second prebuilt-xxx-sd-xx.img, we use it to make a bootable SD card (imagine the old time people use a WinPE USB stick to boot and backup Windows!).

Prepare a Bootable SD Card

On your Windows PC, we open SD_Firmware_Tool.exe to process 1,2,3,4,5 steps to create a bootable SD card.

You need to download the Chipsee prebuilt image as we mentioned earlier. Find the one that fits your screen size in **Chipsee prebuilt image** page.

Once the SD card is flashed, Windows will show a warning to let you format the unrecognized partition, **ignore or cancel** it because the SDDiskTool creates some partitions that Windows doesn't recognize.



Follow the 5 steps on SDDisktool

Backup Your eMMC

Insert this SD card into the SD slot of the Chipsee PC and power it on, the Chipsee PC will boot into the system on the SD card (may need to reboot multiple times to boot from SD card, because of a known bug in the eMMC firmware, we will release a firmware for SD card in the future), we can use this system to backup the whole contents on eMMC rootfs partitions.

Use the way you like to execute the following commands, for example, serial debug or ssh. You can connect a keyboard and mouse to the Chipsee device and run them in the command line as well.

The eMMC rootfs partition is **/dev/mmcb1k0p7**. We will backup the contents in **/dev/mmcb1k0p7**.

```

root@linaro-alip:~# lsblk
NAME                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
mmcblk0             179:0    0 14.6G  0 disk
├─mmcblk0p1         179:1    0   4M  0 part
├─mmcblk0p2         179:2    0   4M  0 part
├─mmcblk0p3         179:3    0   4M  0 part
├─mmcblk0p4         179:4    0  64M  0 part
├─mmcblk0p5         179:5    0 128M  0 part
├─mmcblk0p6         179:6    0  32M  0 part
└─mmcblk0p7         179:7    0 14.3G  0 part
mmcblk0boot0        179:32    0   4M  1 disk
mmcblk0boot1        179:64    0   4M  1 disk
mmcblk1             179:96    0 29.7G  0 disk
├─mmcblk1p1         179:97    0   4M  0 part
├─mmcblk1p2         179:98    0   4M  0 part
├─mmcblk1p3         179:99    0   4M  0 part
├─mmcblk1p4         179:100   0  64M  0 part
├─mmcblk1p5         179:101   0 128M  0 part
├─mmcblk1p6         179:102   0  32M  0 part
└─mmcblk1p7         179:103   0 29.5G  0 part /

```

1 eMMC root partition

2 SD root partition

eMMC rootfs partition is **/dev/mmcblk0p7**

```

$ sudo su
# export ROOTFS_DEV=/dev/mmcblk0p7
# mkdir /mnt/backuprootfs
# mount $ROOTFS_DEV /mnt/backuprootfs/
# cd /mnt/

// sync would take an hour or more depending on the files in your system
# tar --numeric-owner -jcvpf backuprootfs.tar.bz2 backuprootfs && sync
# umount /mnt/backuprootfs

```

Now we have obtained the backup rootfs **backuprootfs.tar.bz2** in the SD card partition

Generate New Image File

Poweroff the Chipsee PC. Put the SD card into your Linux HOST PC (or virtual machine).

You should find a **/dev/sdX** in your Linux system, for example **/dev/sdb**, which is this SD card, **you should use your actual /dev/sdX here**, if you don't know which sdX is it, check with `df -h` and see which one's size is most likely your SD card.

Now we mount **/dev/sdb7** to find **backuprootfs.tar.bz2**

```
# mount /dev/sdb7 /mnt/
```

It will be in **/mnt/mnt/backuprootfs.tar.bz2**, we will copy it out to our Linux PC later.

Run the following command to generate a new *.img* file. Make sure you have at least 25GB free space on your Linux PC, the process produces a lot of intermediate files.

```
$ sudo su
# git clone https://gitee.com/chipsee_admin/rk_pack_tools.git
# cd rk_pack_tools
# git checkout r510-rk3399

// copy the Chipsee prebuilt img file to this directory
# cp prebuilt-xxx.img .
# ./cs-unpack.sh prebuilt-xxx.img

// copy your backup rootfs from SD card to this directory
# cp /mnt/mnt/backuprootfs.tar.bz2 .

// generate rootfs.img file from backuprootfs.tar.bz2
# ./cs-mkrootfs.sh

// generate new img file
# ./cs-pack.sh prebuilt-new-xxx.img
```



Warning

If you see *checksum miss match error* or *Error:<AddFile> write file failed,err=28*, check your harddisk and make sure you have enough free space.

Now you have obtained your new img file *prebuilt-new-xxx.img* in the current folder, use this img file to flash other devices.

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